

TrES-1b

R-1518

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_220607 · created 2026-08-01T16:46:18+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459734.7918 +/- 0.0026 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.044 +/- 0.054
Transit depth (Rp/Rs)^2	0.2 +/- 0.48 [%]
Orbital Inclination (inc)	88.64 +/- 1.47
Airmass coefficient 1 (a1)	0.0091 +/- 0.0037
Airmass coefficient 2 (a2)	3.0 +/- 1.17
Scatter in the residuals of the lightcurve fit is	28.19 %
Best Comparison Star	#2 - [74, 281]
Optimal Aperture	7.16
Optimal Annulus	10.74
Transit Duration (day)	0.0 +/- 0.0

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.044 $\pm$ 0.054 vs 0.1358 (deviation 1.7 σ)
Duration fit vs published	0.0 d vs 0.1042 d (deviation inf σ)
Mid-time O-C	-4357.9 min
Depth SNR	0.8
Residual scatter	28.19 %

Light curve

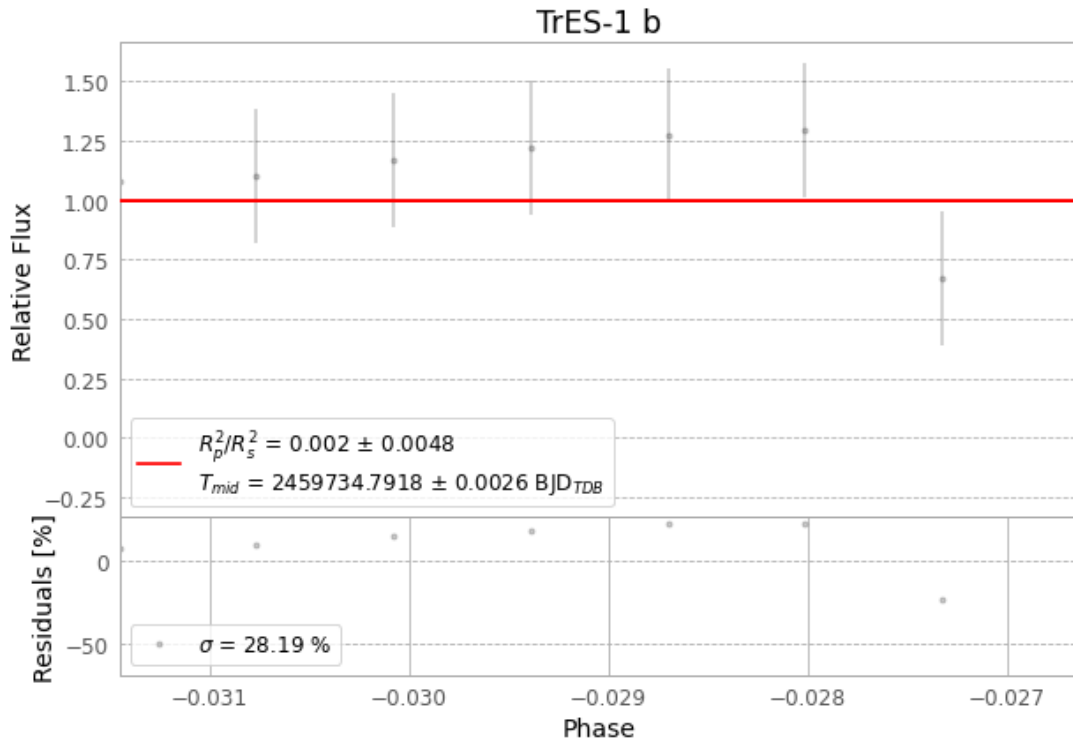


Figure 1 — FinalLightCurve_TrES-1 b_2022-06-06.png (SHA-256 09eb42e403d259fe...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [29, 434], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 006b3d3df1e1e7f2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

91 FITS frames (60 MB), TRES-1220607052113.FITS → TRES-1220607095113.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-220607.log (14ec3f74059e...), FinalLightCurve_TrES-1 b_2022-06-06.png (09eb42e403d2...), FinalLightCurve_TrES-1 b_2022-06-06.csv (817135c07ee3...)

Compute resources

Wall time 390.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 16:46Z.